Aadi Samineni

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Education

University of Michigan — **Ann Arbor, MI**
B.S. in Industrial and Operations Engineering — *Expected Graduation: Dec. 2026*
Projected Master's in IOE (Master's Program Area: Financial Engineering) — *Expected Graduation: Dec. 2027*
GPA: 3.76 / 4.00 | Track in Financial Engineering | Minor in Computer Science and Math
Coursework: Derivative Instruments, Markov Decision Process, Regression Analysis, Probability Theory, Object Oriented Programming, Mathematics of Machine Learning, Data Analytics, Cryptography Theory, Linear Algebra, Calculus I/II

Experience

Options Analyst Intern — **Virtual**
Boutique Hedge Fund — *May 2025 – Sept. 2025*
- Built a PyQt analytics dashboard for signals, positions, and PnL to support backtesting and model debugging
- Engineered and deployed a Random Forest classification trading model to generate signals from open-tick stock data; performed advanced feature engineering and statistical validation, achieving 45% monthly returns in backtesting
- Collaborated with a team to create a supervised logistic regression model through Scikit-learn that hedges on tech stock short straddles, leading to an average PNL increase of 25%

RWA Team Lead — **Ann Arbor, MI**
Michigan Blockchain Investment Team — *Jan 2024 – Present*
- Organized the largest Fintech conference in the Midwest with 1000+ attendees, 20+ companies, recognized by Forbes
- Executed 15-minute Bitcoin futures trades on Polymarket with a trading team, generating 33% returns over two weeks
- Led a team to pitch a tokenized real-world asset for ONDO, analyzing the growing international interest in U.S. debt markets, institutional backing, and dispersing concerns about token dilution, projecting a 9x growth over the next 6 years

Research Assistant — **Ann Arbor, MI**
University of Michigan Department of Financial Engineering — *Oct 2024 – Present*
- Conducted research on RSU and AV valuation, formulating PDE/HJB-based value functions and training a simple neural network to approximate derivatives of the value function, enabling analysis of optimal conversion and exercise policies
- Implemented real-time implied volatility estimation by hand-coding Newton–Raphson inversion for Black–Scholes

Personal Projects

Prediction Markets Algorithms | *Python, Websockets, Pandas,*
- Built a low-latency WebSocket client to ingest real-time Kalshi market data and stream updates into the trading/feature pipeline, reducing data latency and enabling near real-time signal generation
- Engineered a low-latency quantitative trading system exploiting order book imbalances in Kalshi basketball markets, generating 1% edge per game and achieving 40% ROI over one month with 67% hit rate
- Designed a feature engineering pipeline for order book and event-driven data, and assessed Mamba-style state-space models versus lighter-weight alternatives based on predictive value, stability, and deployment feasibility
- Developed an interactive live-feed GUI for monitoring Kalshi and Polymarket basketball markets in real time

Algorithmic Trading Competition – IMC Prosperity | *Python, Logistic Regression, Fourier Transform*
- Engineered a robust volatility solution with real-time crooked smile implied volatility curve fitting, delta/gamma hedging, and adaptive position scaling to optimize risk-adjusted returns in unpredictable, rapidly evolving market conditions
- Performed exploratory data analysis and feature engineering on historical price movements, applying logistic regression and fast fourier models to extract predictive trading signals and drive iterative strategy refinement
- Ranked in the top 5% of over 13,000+ competitors by fine-tuning synthetic basket arbitrage execution under liquidity constraints, position limits and auction-based trade fulfillment capitalizing on bid-ask spreads

March Madness Bracket Producer | *Python, Beautiful Soup, Selenium*
- Engineered a bracket generation engine to simulate hundreds of March Madness outcomes using probabilistic modeling and data-driven team weightings, optimizing prediction quality across different seeding scenarios
- Automated bulk entry submission by scripting Selenium-based account creation and bracket uploads to the CBS Bracket Challenge platform, ensuring scalable deployment and consistent entry selection
- Outperformed baseline scores by an average of 30%, validating model efficacy against CBS participant benchmarks

Technical Skills

Languages: Python, C++, Matlab, SQL
Technologies: NumPy, Pandas, SQLite, Scikit-learn, Selenium, Excel, Websocket, Matplotlib, Git, PyQT, PyTorch
Awards: JPM Case Competition: 1st, P&G Case Competition: 1st, Franklin Templeton Pitch Competition: Semi-finalist
Interests: Guitar, Basketball, Weightlifting, Football, Running, Trumpet, Cooking